

A cubic girth bound for even cycle intervals

A graph is *cubic* if every vertex has degree three. Its *girth* is the length of its shortest simple cycle. All logarithms denoted by $\log_2$ are to base two.

Theorem. *Let G be a finite simple cubic graph on $n \geq 2^{64}$ vertices, and let g be its girth. Suppose that*

$$g \geq \lceil 4 \log_2(\log_2 n) + 4 \log_2(\log_2(\log_2 n)) + 22 \rceil. \quad (1)$$

Set

$$t = \log_2 n, \quad \ell = \log_2 t, \quad H = 2\ell + 2 \log_2 \ell + 10. \quad (2)$$

Then, for every even integer q satisfying

$$tH \leq q \leq 2tH, \quad (3)$$

the graph G contains a simple cycle of length exactly q .

Proof. Fix an even integer q satisfying (3). Write $V = V(G)$, let A be the adjacency matrix of G , and put $m = |E(G)| = 3n/2$. A finite graph of minimum degree at least two contains a cycle, so g is finite.

1. Walk counts, the girth moment, and repeated vertices

A walk $(v_0, v_1, \dots, v_r)$ is *internally nonbacktracking* if $v_{j-1} \neq v_{j+1}$ for $1 \leq j \leq r-1$. If $v_r = v_0$, this definition places no restriction at the closing junction: it may happen that $v_{r-1} = v_1$. Let $f_r(v)$ be the number of these length- r return walks based at v .

Let $M_r(u, v)$ count internally nonbacktracking walks of length r from u to v . Then

$$M_1 = A, \quad M_2 = A^2 - 3I, \quad M_r = M_{r-1}A - 2M_{r-2} \quad (r \geq 3).$$

Indeed, $M_{r-1}A$ counts extensions by one edge. An extension backtracks at its last step precisely when it appends the two-edge segment w, z, w to a length- $(r-2)$ walk ending at w . For $r \geq 3$ that shorter walk has a last edge, and there are exactly two choices of z other than its preceding vertex. For $r = 2$, the three immediate returns are subtracted instead. Consequently $M_r = P_r(A)$, where

$$\begin{aligned} P_1(x) &= x, & P_2(x) &= x^2 - 3, \\ P_r(x) &= xP_{r-1}(x) - 2P_{r-2}(x) & (r \geq 3). \end{aligned} \quad (4)$$

In particular,

$$f_r(v) = (P_r(A))_{vv} \geq 0. \quad (5)$$

Put

$$h = \left\lfloor \frac{g-1}{2} \right\rfloor, \quad \text{so that } h \geq 1 \text{ and } 2h < g. \quad (6)$$

Every internally nonbacktracking walk of length h is simple. To see this, take two occurrences of a repeated vertex with the smallest positive separation. The intervening segment is a simple cycle: its length cannot be one because G has no loops, or two because the walk does not backtrack. Such a cycle would have length at most $h < g$. Moreover, two distinct simple paths of length h with the same endpoints have a cycle in their union, of length at most $2h < g$. Therefore there is at most one internally nonbacktracking walk of length h between any ordered pair of vertices.

For $r \geq g$, split a length- r return walk at v after $r-h$ steps. There are exactly $3 \cdot 2^{r-h-1}$ possible internally nonbacktracking prefixes, and for each prefix there is at most one length- h

suffix returning to v . Some such concatenations may be inadmissible; this only decreases the count. Thus

$$f_r(v) \leq G_r := \frac{3}{2} 2^{r-h} \quad (r \geq g). \quad (7)$$

The matrix $P_h(A)$ is symmetric, has entries in $\{0, 1\}$, and has exactly $3 \cdot 2^{h-1}$ ones in each row, since this is the number of length- h nonbacktracking walks starting at a given vertex. It follows that, at every vertex,

$$(P_h(A)^2)_{vv} = \sum_{u \in V} (P_h(A)_{vu})^2 = 3 \cdot 2^{h-1}. \quad (8)$$

Let $\vec{E}$ be the set of directed edges, and write $\bar{e}$ for the reverse of $e \in \vec{E}$. The nonbacktracking matrix B , indexed by $\vec{E}$, is defined by

$$B_{e,f} = \begin{cases} 1, & \text{if the head of } e \text{ is the tail of } f \text{ and } f \neq \bar{e}, \\ 0, & \text{otherwise.} \end{cases}$$

Thus $T_q := \text{tr}(B^q)$ counts ordered tuples $(e_0, \dots, e_{q-1})$ satisfying the nonbacktracking transition rule also from e_{q-1} to e_0 . These are rooted, oriented, cyclically nonbacktracking closed walks: the starting directed edge is distinguished and is summed over all of $\vec{E}$.

Write the corresponding vertex sequence as $(v_0, \dots, v_{q-1}, v_q)$, where $v_q = v_0$. Let D_q count such tuples for which a vertex is repeated among $v_0, \dots, v_{q-1}$; the mandatory equality $v_q = v_0$ is excluded from this definition. If c_q is the number of undirected simple cycles of length q , then

$$T_q - D_q = 2q c_q. \quad (9)$$

Here a simple cycle has exactly q choices of starting vertex and two orientations.

Every walk counted by D_q has two distinct numerical positions carrying the same vertex. Cutting at those positions produces two positive-length internally nonbacktracking return walks. Each such return walk contains a simple cycle, by the minimal-repetition argument above; hence each segment has length at least g . One of the two segments has length $r \leq \lfloor q/2 \rfloor$. Consequently the original cyclic word has a numerical position $p \in \{0, \dots, q-1\}$ with $v_p = v_{p+r}$, where indices are taken modulo q and $g \leq r \leq \lfloor q/2 \rfloor$.

For fixed p, r , and $v = v_p$, reading the two segments in the original orientation gives a pair counted by $f_r(v)f_{q-r}(v)$. The pair and the numerical position p reconstruct the original rooted tuple, so this map is injective for fixed p and r . Summing over possible marks therefore gives the upper bound

$$D_q \leq q \sum_{r=g}^{\lfloor q/2 \rfloor} \sum_{v \in V} f_r(v)f_{q-r}(v). \quad (10)$$

Multiple admissible marks and inadmissible concatenations cause only overcounting. In particular, no division by a rotation orbit, and no assumption of aperiodicity, is involved. If the range of r is empty, then $D_q = 0$.

2. The nonbacktracking determinant and polynomial bounds

We derive the determinant identity needed below. Let U and W be the $n \times 2m$ tail and head incidence matrices, respectively, and let J be the $2m \times 2m$ matrix reversing directed edges. Then

$$B = W^\top U - J, \quad J^2 = I, \quad UW^\top = A, \quad UJW^\top = 3I.$$

Each undirected edge gives a two-by-two block of J , so $\det(I + uJ) = (1 - u^2)^m$. For $u \neq \pm 1$, the identity $(I + uJ)^{-1} = (I - uJ)/(1 - u^2)$ and the rectangular determinant identity $\det(I + XY) = \det(I + YX)$ give

$$\begin{aligned}\det(I - uB) &= \det(I + uJ) \det(I - uU(I + uJ)^{-1}W^\top) \\ &= (1 - u^2)^m \det\left(I - \frac{u}{1 - u^2}(A - 3uI)\right) \\ &= (1 - u^2)^{m-n} \det(I - uA + 2u^2I).\end{aligned}\tag{11}$$

The rectangular determinant identity itself follows by evaluating $\det\begin{pmatrix} I & X \\ -Y & I \end{pmatrix}$ using either diagonal block. Since both sides of (11) are polynomials in u , the identity holds for all u .

Let $\lambda_1, \dots, \lambda_n$ be the adjacency eigenvalues, with multiplicity. The matrix A is real symmetric, $A\mathbf{1} = 3\mathbf{1}$, and all its eigenvalues lie in $[-3, 3]$. The last assertion follows, for example, by applying an eigenvector equation at a coordinate of maximum absolute value. Equation (11) gives

$$\det(zI - B) = (z^2 - 1)^{m-n} \prod_{i=1}^n (z^2 - \lambda_i z + 2).\tag{12}$$

For $|\lambda_i| > 2\sqrt{2}$, define

$$\beta_i = \frac{|\lambda_i| + \sqrt{\lambda_i^2 - 8}}{2}, \quad \gamma_i = \frac{2}{\beta_i}.\tag{13}$$

Then $\sqrt{2} < \beta_i \leq 2$. The two roots associated with λ_i are β_i, γ_i when $\lambda_i > 0$, and $-\beta_i, -\gamma_i$ when $\lambda_i < 0$. For $|\lambda_i| \leq 2\sqrt{2}$, both roots have absolute value $\sqrt{2}$, including the repeated-root cases.

Choose, for now, any two cutoffs satisfying

$$\sqrt{2} < \beta_c < \beta_* < 2, \quad C = \frac{\beta_c^2 - 1}{\beta_c^2 - 2}.\tag{14}$$

We next prove the following estimates for every integer $r \geq 1$ and every real λ in the indicated intervals:

$$|P_r(\lambda_i)| \leq C\beta_i^r \quad \text{if } |\lambda_i| > 2\sqrt{2} \text{ and } \beta_i > \beta_c,\tag{15}$$

$$|P_r(\lambda)| \leq Cb^r \quad \text{if } |\lambda| \leq b + 2/b, \quad b \in \{\beta_c, \beta_*\},\tag{16}$$

$$|P_r(\lambda_i)| \geq \frac{3}{2}\beta_i^r \quad \text{if } |\lambda_i| > 2\sqrt{2}.\tag{17}$$

For $\lambda = \beta + 2/\beta$ with $\sqrt{2} < \beta \leq 2$, solving (4) gives

$$P_r(\lambda) = a_\beta \beta^r - b_\beta (2/\beta)^r, \quad a_\beta = \frac{\beta^2 - 1}{\beta^2 - 2}, \quad b_\beta = \frac{4 - \beta^2}{2(\beta^2 - 2)}.\tag{18}$$

This formula is verified at $r = 1, 2$, and both sides satisfy the same recurrence for $r \geq 3$. Here $b_\beta \geq 0$, $a_\beta - b_\beta = 3/2$, and

$$\frac{d}{d\beta} a_\beta = -\frac{2\beta}{(\beta^2 - 2)^2} < 0.$$

Since $2/\beta \leq \beta$, it follows that

$$\frac{3}{2}\beta^r \leq P_r(\beta + 2/\beta) \leq a_\beta \beta^r.\tag{19}$$

Also $P_r(-x) = (-1)^r P_r(x)$ by (4). These observations prove (15) and (17).

To include the whole low spectral interval, the same recurrence gives, for real or complex θ ,

$$P_r(2\sqrt{2}\cos\theta) = 2^{r/2} \left(e^{ir\theta} + e^{-ir\theta} + \frac{1}{2} \sum_{j=1}^{r-1} e^{i(r-2j)\theta} \right), \quad (20)$$

with the sum empty when $r = 1$. This identity follows by checking $r = 1, 2$ and the recurrence. If θ is real, the absolute value of the right-hand side is at most its value at $\theta = 0$, namely $2^{r/2}(r+3)/2$. If $\theta = iu$ with $u \geq 0$, pair opposite exponents in (20). The result is a positive linear combination of functions $\cosh(ku)$ with nonnegative integers k , and hence is nondecreasing in u . Since

$$b + 2/b = 2\sqrt{2} \cosh(\ln(b/\sqrt{2})) \quad (b > \sqrt{2}),$$

parity and the two preceding bounds imply

$$|P_r(\lambda)| \leq P_r(b + 2/b) \leq a_b b^r \quad \text{whenever } |\lambda| \leq b + 2/b.$$

For $b = \beta_c$ we have $a_b = C$, and for $b = \beta_*$ we have $a_b < C$. This proves (16), including the endpoints $\lambda = \pm 2\sqrt{2}$.

Take a real orthonormal eigenbasis $\phi_1, \dots, \phi_n$ of A , and set

$$w_i(v) = \phi_i(v)^2, \quad \sum_{i=1}^n w_i(v) = 1, \quad \sum_{v \in V} w_i(v) = 1. \quad (21)$$

Define the index sets

$$\mathcal{I}_c = \{i : |\lambda_i| > 2\sqrt{2}, \beta_i > \beta_c\}, \quad \mathcal{I}_* = \{i : |\lambda_i| > 2\sqrt{2}, \beta_i > \beta_*\},$$

so that $\mathcal{I}_* \subseteq \mathcal{I}_c$, and write

$$H_r(v) = \sum_{i \in \mathcal{I}_c} w_i(v) \beta_i^r, \quad H_r^*(v) = \sum_{i \in \mathcal{I}_*} w_i(v) \beta_i^r, \quad S_q^* = \sum_{i \in \mathcal{I}_*} \beta_i^q. \quad (22)$$

The function $b \mapsto b + 2/b$ is increasing for $b > \sqrt{2}$. Thus (5), the spectral decomposition of $P_r(A)$, and (15)–(16) imply

$$f_r(v) \leq C H_r(v) + C \beta_c^r, \quad f_r(v) \leq C H_r^*(v) + C \beta_*^r. \quad (23)$$

For $r \geq g$, put

$$L_r = \min\{G_r, C \beta_c^r\}. \quad (24)$$

Combining the first inequality in (23) with (7) gives

$$f_r(v) \leq C H_r(v) + L_r \quad (r \geq g). \quad (25)$$

Indeed, for nonnegative a, b, c , one has $\min\{c, a + b\} \leq a + \min\{c, b\}$.

3. A local correlation estimate and lower bounds for the trace

By (17), (8), and the spectral decomposition,

$$\begin{aligned} m_h(v) &:= \sum_{i \in \mathcal{I}_c} w_i(v) \beta_i^{2h} \\ &\leq \frac{4}{9} \sum_{i=1}^n w_i(v) P_h(\lambda_i)^2 = \frac{4}{9} (P_h(A)^2)_{vv} = \frac{2}{3} 2^h. \end{aligned} \quad (26)$$

We claim that, whenever $r + s = q$ and $r, s \geq g$,

$$\sum_{v \in V} H_r(v) H_s^*(v) \leq \frac{2}{3} \left(\frac{2}{\beta_*^2} \right)^h S_q^*. \quad (27)$$

Fix a vertex v . On the index set $\mathcal{I}_c$, set

$$t_i = w_i(v) \beta_i^{2h}, \quad a_i = \beta_i^{r-2h}, \quad b_i = \beta_i^{s-2h} \mathbf{1}_{\{\beta_i > \beta_*\}}.$$

Since $r, s \geq g > 2h$, the quantities a_i and b_i are nondecreasing functions of β_i . The jump in the latter function at β_* is upward. Therefore

$$\begin{aligned} & \left(\sum_{i \in \mathcal{I}_c} t_i \right) \left(\sum_{i \in \mathcal{I}_c} t_i a_i b_i \right) - \left(\sum_{i \in \mathcal{I}_c} t_i a_i \right) \left(\sum_{i \in \mathcal{I}_c} t_i b_i \right) \\ &= \frac{1}{2} \sum_{i, j \in \mathcal{I}_c} t_i t_j (a_i - a_j)(b_i - b_j) \geq 0. \end{aligned}$$

There is no division by $\sum_i t_i$, so this also covers the case $m_h(v) = 0$. Substituting the definitions gives

$$H_r(v) H_s^*(v) \leq m_h(v) \sum_{i \in \mathcal{I}_*} w_i(v) \beta_i^{q-2h}.$$

Summing over v and using (26) and (21), we obtain

$$\begin{aligned} \sum_{v \in V} H_r(v) H_s^*(v) &\leq \frac{2}{3} 2^h \sum_{i \in \mathcal{I}_*} \beta_i^{q-2h} \\ &\leq \frac{2}{3} 2^h \beta_*^{-2h} \sum_{i \in \mathcal{I}_*} \beta_i^q, \end{aligned}$$

which is (27).

We next control T_q . The trace of a matrix power is the sum of the corresponding powers of its eigenvalues, counted with algebraic multiplicity. This follows by triangularizing the matrix over $\mathbb{C}$ and does not require diagonalizability. In (12), each of the $m - n$ additional roots $+1$ and each of the $m - n$ additional roots -1 contributes 1 to T_q , since q is even. Their total contribution is $2(m - n) = n$. For $|\lambda_i| > 2\sqrt{2}$, the associated pair contributes $\beta_i^q + \gamma_i^q \geq 0$, and at least β_i^q when $i \in \mathcal{I}_*$. For $|\lambda_i| \leq 2\sqrt{2}$, the sum of the q th powers is real and at least $-2 \cdot 2^{q/2}$, by the triangle inequality. Finally, the adjacency eigenvalue 3 provides a root 2, which is included in S_q^* because $\beta_* < 2$. Consequently

$$T_q \geq S_q^* - 2n2^{q/2}, \quad S_q^* \geq 2^q, \quad T_q \geq 2^q - 2n2^{q/2}. \quad (28)$$

Put

$$\eta_q = 2n2^{-q/2}. \quad (29)$$

Whenever $\eta_q < 1$, we have

$$T_q \geq (1 - \eta_q)2^q > 0, \quad \frac{S_q^*}{T_q} \leq 1 + \frac{\eta_q 2^q}{T_q} \leq \frac{1}{1 - \eta_q}. \quad (30)$$

4. A finite sufficient condition for a simple cycle

Define the convergent series

$$\begin{aligned} W_g &:= \sum_{r=g}^{\infty} L_r \beta_*^{-r} \\ &= \sum_{r=g}^{\infty} \min \left\{ \frac{3}{2} 2^{-h} \left(\frac{2}{\beta_*} \right)^r, C \left(\frac{\beta_c}{\beta_*} \right)^r \right\}. \end{aligned} \quad (31)$$

Convergence follows from $\beta_c/\beta_* < 1$ by bounding the minimum by its second entry. We will show that, provided $\eta_q < 1$, it suffices to verify

$$\frac{C^2}{3} q^2 \left(\frac{2}{\beta_*^2} \right)^h + CqW_g + \frac{3C}{4} nq^2 2^{-h} \left(\frac{\beta_*}{2} \right)^{q/2} < 1 - \eta_q. \quad (32)$$

For a split length $g \leq r \leq \lfloor q/2 \rfloor$, put $s = q - r$. Since $f_r(v) \geq 0$, the second bound in (23) first implies

$$f_r(v)f_s(v) \leq Cf_r(v)H_s^*(v) + Cf_r(v)\beta_*^s.$$

Use (25) on the first term and (7) on the second. This gives

$$f_r(v)f_s(v) \leq C^2 H_r(v)H_s^*(v) + CL_r H_s^*(v) + CG_r \beta_*^s. \quad (33)$$

The first term is controlled after summing over v by (27). For the second, (21) gives

$$L_r \sum_{v \in V} H_s^*(v) = L_r \sum_{i \in \mathcal{I}_*} \beta_i^{q-r} \leq L_r \beta_*^{-r} S_q^*. \quad (34)$$

For the third, $s \geq q/2$ and $0 < \beta_*/2 < 1$ imply

$$G_r \beta_*^s = \frac{3}{2} 2^{q-h} \left(\frac{\beta_*}{2} \right)^s \leq \frac{3}{2} 2^{q-h} \left(\frac{\beta_*}{2} \right)^{q/2}. \quad (35)$$

There are at most $q/2$ split lengths. Substituting (33)–(35) into (10), and extending the middle sum to the series (31), yields

$$\begin{aligned} D_q &\leq \left[\frac{C^2}{3} q^2 \left(\frac{2}{\beta_*^2} \right)^h + CqW_g \right] S_q^* \\ &\quad + \frac{3C}{4} nq^2 2^{q-h} \left(\frac{\beta_*}{2} \right)^{q/2}. \end{aligned} \quad (36)$$

Indeed, the coefficients of the first and last terms are, respectively, $q(q/2)C^2(2/3) = C^2 q^2/3$ and $q(q/2)C(3/2) = 3Cq^2/4$. Divide (36) by $T_q > 0$ and use (30). The result is

$$\frac{D_q}{T_q} \leq \frac{1}{1 - \eta_q} \left[\frac{C^2}{3} q^2 \left(\frac{2}{\beta_*^2} \right)^h + CqW_g + \frac{3C}{4} nq^2 2^{-h} \left(\frac{\beta_*}{2} \right)^{q/2} \right]. \quad (37)$$

Thus (32) implies $D_q < T_q$, and (9) then gives $c_q > 0$.

5. Adaptive parameters and exact verification of the certificate

Since $n \geq 2^{64}$, $t \geq 64$ and $\ell \geq 6$. Moreover,

$$4 \log_2(\log_2 n) + 4 \log_2(\log_2(\log_2 n)) + 22 = 2H + 2.$$

For even g , we have $h = (g - 2)/2$, and for odd g , we have $h = (g - 1)/2$. In either case $h \geq (g - 2)/2$, and hence

$$h \geq \frac{g - 2}{2} \geq H. \quad (38)$$

Also $6^2 > 2^5$, so $\log_2 6 > 5/2$, and consequently

$$H \geq 22 + 2 \log_2 6 > 27. \quad (39)$$

Choose

$$\begin{aligned} \beta_c &= 2^{3/4}, & C &= \frac{3 + \sqrt{2}}{2} < \frac{9}{4}, \\ x = \frac{q}{tH} &\in [1, 2], & \delta &= \frac{2t}{q} = \frac{2}{xH}, & \beta_* &= 2^{1-\delta}. \end{aligned} \quad (40)$$

The bound on C follows from $\sqrt{2} < 3/2$. Equations (39) and (40) give $0 < \delta \leq 2/H < 2/27 < 1/12$, so the cutoff ordering (14) holds. In addition,

$$n \left(\frac{\beta_*}{2} \right)^{q/2} = 2^{t-\delta q/2} = 1. \quad (41)$$

Bounding the series W_g . Let

$$d = \frac{1}{4} - \delta > \frac{1}{6}, \quad R = 4h + 3.$$

Since $2/\beta_* = 2^\delta$ and $\beta_c/\beta_* = 2^{-d}$, use the first entry of the minimum in (31) for $0 \leq r < R$ and the second for $r \geq R$. Extending the starting range from g to 0 only adds nonnegative terms. Thus

$$\begin{aligned} W_g &\leq \frac{3}{2} 2^{-h} \sum_{r=0}^{R-1} 2^{\delta r} + C \sum_{r=R}^{\infty} 2^{-dr} \\ &= \frac{3}{2} 2^{-h} \frac{2^{\delta R} - 1}{2^\delta - 1} + \frac{C 2^{-dR}}{1 - 2^{-d}} \\ &\leq 2^{-(1-4\delta)h} \left[\frac{(3/2)2^{3\delta}}{2^\delta - 1} + \frac{C 2^{-3d}}{1 - 2^{-d}} \right]. \end{aligned} \quad (42)$$

Here $4d = 1 - 4\delta$ was used in the last line.

For explicit bounds, strict convexity of $u \mapsto 1/u$ gives

$$\ln 2 = \int_1^2 \frac{du}{u} > \int_1^2 \left(\frac{4}{3} - \frac{4}{9}u \right) du = \frac{2}{3}.$$

Also $(6/5)^4 = 1296/625 > 2$, so $2^{1/4} < 6/5$. Using $2^a - 1 \geq a \ln 2$ for $a > 0$ (which follows from $e^y \geq 1 + y$), we obtain

$$\frac{(3/2)2^{3\delta}}{2^\delta - 1} < \frac{(3/2)(6/5)}{(2/3)\delta} = \frac{27}{10\delta}. \quad (43)$$

For the other term, retaining its numerator gives

$$\frac{C 2^{-3d}}{1 - 2^{-d}} = \frac{C 2^{-2d}}{2^d - 1} < \frac{C}{d \ln 2} < \frac{9/4}{(1/6)(2/3)} = \frac{81}{4}. \quad (44)$$

Since $1 - 4\delta > 0$, (38) allows h to be replaced by H in the decreasing exponential in (42). Because $4\delta H = 8/x$, the last three estimates yield

$$W_g < \left(\frac{27xH}{20} + \frac{81}{4} \right) 2^{-H+8/x}. \quad (45)$$

Reducing to two endpoints. Denote the left-hand side of (32) by $\mathcal{F}$. Set

$$Z = t^2 H^2 2^{-H} = 2^{-10} \left(\frac{H}{\ell} \right)^2, \quad (46)$$

where the equality uses $t = 2^\ell$ and $2^H = 2^{10}t^2\ell^2$. Since $2/\beta_*^2 = 2^{-1+2\delta} < 1$, equations (38), (40), (41), and (45) give

$$\begin{aligned}\frac{C^2}{3}q^2\left(\frac{2}{\beta_*^2}\right)^h &< Z\frac{27}{16}x^22^{4/x}, \\ CqW_g &< Z\frac{9}{4t}\left(\frac{27}{20}x^2 + \frac{81}{4H}x\right)2^{8/x}, \\ \frac{3C}{4}nq^22^{-h}\left(\frac{\beta_*}{2}\right)^{q/2} &< Z\frac{27}{16}x^2.\end{aligned}$$

In particular,

$$\begin{aligned}\mathcal{F} < \Phi(x) := Z\left[\frac{27}{16}x^22^{4/x} + \frac{27}{16}x^2\right. \\ \left. + \frac{9}{4t}\left(\frac{27}{20}x^2 + \frac{81}{4H}x\right)2^{8/x}\right].\end{aligned}\tag{47}$$

In analyzing Φ , the quantities t, H, Z are held fixed. For every $c > 0$ and $x > 0$,

$$\begin{aligned}\frac{d^2}{dx^2}(x^2e^{c/x}) &= e^{c/x}\left[\left(\frac{c}{x} - 1\right)^2 + 1\right] > 0, \\ \frac{d^2}{dx^2}(xe^{c/x}) &= \frac{c^2}{x^3}e^{c/x} > 0.\end{aligned}\tag{48}$$

Thus Φ is convex: its terms are positive multiples of x^2 , $x^2e^{c/x}$, or $xe^{c/x}$, with $c = 4\ln 2$ or $8\ln 2$. It follows that, for $1 \leq x \leq 2$,

$$\Phi(x) \leq (2-x)\Phi(1) + (x-1)\Phi(2) \leq \max\{\Phi(1), \Phi(2)\}.$$

Direct substitution gives the two endpoint values

$$\begin{aligned}E_1 := \Phi(1) &= Z\left[\frac{459}{16} + \frac{576}{t}\left(\frac{27}{20} + \frac{81}{4H}\right)\right], \\ E_2 := \Phi(2) &= Z\left[\frac{135}{4} + \frac{36}{t}\left(\frac{27}{5} + \frac{81}{2H}\right)\right].\end{aligned}\tag{49}$$

We now regard $t = 2^\ell$ and $H = 2\ell + 2\log_2 \ell + 10$ as functions of the real variable $\ell \geq 6$. Both t and H increase, whereas

$$\frac{d}{d\ell}\left(\frac{H}{\ell}\right) = \frac{2/\ln 2 - 2\log_2 \ell - 10}{\ell^2} < 0.\tag{50}$$

For the strict inequality, use $2/\ln 2 < 3$ and $\ell \geq 6$. Hence Z decreases, and each positive bracket in (49) also decreases. Therefore both endpoint values are largest at $\ell = 6$.

At this endpoint, $t = 64$ and

$$H_0 := H|_{\ell=6} = 22 + 2\log_2 6.$$

The exact comparison $6^5 = 7776 < 8192 = 2^{13}$ gives $\log_2 6 < 13/5$, so $H_0 < 136/5$. Using $Z = H_0^2/36864$ at $\ell = 6$, equations (49) reduce to

$$E_1|_{\ell=6} = \frac{(3267/80)H_0^2 + (729/4)H_0}{36864} < \frac{244239}{256000} < \frac{31}{32},\tag{51}$$

$$E_2|_{\ell=6} = \frac{(2943/80)H_0^2 + (729/32)H_0}{36864} < \frac{1546473}{2048000} < \frac{31}{32}.\tag{52}$$

The displayed rational upper bounds are obtained by replacing H_0 with $136/5$ in the preceding expressions, whose coefficients are positive. The last comparisons can be checked without rounding:

$$\frac{31}{32} - \frac{244239}{256000} = \frac{3761}{256000} > 0, \quad \frac{31}{32} - \frac{1546473}{2048000} = \frac{437527}{2048000} > 0.$$

Consequently, for every $\ell \geq 6$ and $x \in [1, 2]$,

$$\mathcal{F} < \Phi(x) \leq \max\{E_1, E_2\} < \frac{31}{32}. \quad (53)$$

Finally, (3) and $H > 27$ give

$$\eta_q = 2n2^{-q/2} \leq 2n^{1-H/2} < 2n^{-25/2} \leq 2^{-799} < \frac{1}{32}. \quad (54)$$

In particular $\eta_q < 1$, as required for (30), and (53) implies

$$\mathcal{F} < \frac{31}{32} < 1 - \eta_q.$$

This is precisely (32). Thus (37) gives $D_q < T_q$, and (9) yields

$$c_q = \frac{T_q - D_q}{2q} > 0.$$

Since q was arbitrary, G contains a simple cycle of every even length in (3). □